

William J Vianese

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EDUCATION

WASHINGTON UNIVERSITY IN ST. LOUIS | St. Louis, MO

May 2026

Bachelor of Arts

Majors in Math & Computer Science and Environmental Analysis

- GPA: 3.82/4.00; Dean's List (Spring 2023, Fall 2023, Fall 2024, Fall 2025)

Relevant Coursework

- Data Science, Data Mining, Artificial Intelligence, Analysis of Algorithms, Algorithms for Nonlinear Optimization, Data Structures, Dynamics & Computation, Financial Mathematics, Probability & Statistics, Matrix Algebra, and Applied Math

TECHNICAL SKILLS

- Languages:* Python, Java, R
- Libraries:* Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- Tools:* Google Suite, Microsoft Suite, GitHub, Zoom, Slack

PROJECTS

Ferguson Housing Abandonment Risk Map

- Built an interactive geospatial web application to visualize housing risk for 8,800+ parcels with layered map views, filters, and address search for exploratory analysis
- Used CLIP vision embeddings on Google Street View imagery to classify parcels as occupied, abandoned, or vacant
- Trained a logistic regression model on labeled county records to predict abandonment risk for previously unclassified parcels
- Combined visual and statistical model outputs into a unified risk score and displayed key visual variables to reveal spatial patterns and neighborhood-level trends

Real-Time Multiplayer Monopoly Game

- Designed and built a real-time, full-stack multiplayer game platform with server-side game logic and state synchronization
- Built event-driven turn logic using WebSockets to support concurrent players, auctions, trading, and game state updates
- Implemented persistence and session management to support reconnects and multi-game concurrency
- Deployed the application to a cloud hosting platform and tested with live users

Dimensionality Reduction and Classification

- Applied principal component analysis (PCA) to reduce feature dimensionality and analyze information retention
- Implemented and compared multiple classification models such as logistic regression, k-NN, and decision trees
- Evaluated model performance using accuracy metrics and confusion matrices
- Analyzed tradeoffs between dimensionality reduction, model complexity, and predictive performance

EXPERIENCE

CAMP EMERSON | Hinsdale, MA

May 2022 – August 2025

Roles Held: Data & Systems Manager, Program Director, Wilderness Director, and Counselor

- Progressed across four roles to own and optimize core operational systems supporting 285+ users and 80+ staff, spanning enrollment, scheduling, transportation, billing, and compliance
- Designed and implemented automated workflows and process improvements to improve data consistency, reduce manual operations, and improve system reliability
- Performed data analysis and reporting on enrollment trends, logistical constraints, financial activity, and staffing needs to support leadership decisions

ACTIVITIES

WASHINGTON UNIVERSITY ROWING TEAM | St. Louis, MO

August 2022 - PRESENT

Coach and Coxswain

- Lead workouts, dieting, and recovery in conjunction with motivating and organizing the rowers
- Employ critical and quick-thinking skills to practice effectively on the water and land
- Leverage strong interpersonal skills to promote team cohesion and expand motivation

AWARDS AND RECOGNITIONS

- Selected as ACRA Third Team All American Coxswain 2024
- Selected as ACRA All Regional Coxswain in the Plains Region 2023, 2024